

$$1) a; [J^{\alpha\beta}, J^{\rho\sigma}] = g^{\sigma\beta} J^{\rho\alpha} + g^{\rho\alpha} J^{\sigma\beta} + g^{\rho\beta} J^{\alpha\sigma} + g^{\alpha\sigma} J^{\beta\rho}$$

The commutator is given by

$$[J^{\alpha\beta}, J^{\rho\sigma}] = \underbrace{J^{\alpha\beta} J^{\rho\sigma}}_{\text{I}} - \underbrace{J^{\rho\sigma} J^{\alpha\beta}}_{\text{II}}$$

ACCORDING TO RAMOND $J^{\mu\nu} \equiv i(x^\mu \partial^\nu - x^\nu \partial^\mu)$ IS A

REPRESENTATION; USING IT:

$$\text{I} \quad i^2 (x^\alpha \partial^\beta - x^\beta \partial^\alpha)(x^\rho \partial^\sigma - x^\sigma \partial^\rho) =$$

$$= i^2 [x^\alpha \partial^\rho (x^\rho \partial^\sigma) - x^\alpha \partial^\beta (x^\sigma \partial^\rho) - x^\beta \partial^\alpha (x^\rho \partial^\sigma) + x^\beta \partial^\alpha (x^\sigma \partial^\rho)]$$

$$= i^2 \{ x^\alpha [(\partial^\rho x^\rho) \partial^\sigma + x^\rho (\partial^\rho \partial^\sigma)] - x^\alpha [(\partial^\beta x^\sigma) \partial^\rho + x^\sigma (\partial^\beta \partial^\rho)] - x^\beta [(\partial^\alpha x^\rho) \partial^\sigma + x^\rho (\partial^\alpha \partial^\sigma)] + x^\beta [(\partial^\alpha x^\sigma) \partial^\rho + x^\sigma (\partial^\alpha \partial^\rho)] \}$$

USING THAT $\partial^\mu x^\nu = \frac{\partial}{\partial x_\mu} x^\nu = \frac{\partial}{\partial x_\mu} g^{\nu\mu} x_\mu = g^{\nu\mu}$; THAT YIELDS

$$= i^2 [g^{\rho\beta} x^\alpha \partial^\sigma + x^\alpha x^\rho \partial^\rho \partial^\sigma - g^{\sigma\beta} x^\alpha \partial^\rho - x^\alpha x^\sigma \partial^\beta \partial^\rho - g^{\rho\alpha} x^\beta \partial^\sigma - x^\beta x^\rho \partial^\alpha \partial^\sigma + g^{\sigma\alpha} x^\beta \partial^\rho + x^\beta x^\sigma \partial^\alpha \partial^\rho]$$

$$\text{II} \quad i^2 (x^\rho \partial^\sigma - x^\sigma \partial^\rho)(x^\alpha \partial^\beta - x^\beta \partial^\alpha)$$

$$= i^2 [x^\rho \partial^\sigma (x^\alpha \partial^\beta) - x^\rho \partial^\sigma (x^\beta \partial^\alpha) - x^\sigma \partial^\rho (x^\alpha \partial^\beta) + x^\sigma \partial^\rho (x^\beta \partial^\alpha)]$$

$$= i^2 \{ x^\rho [(\partial^\sigma x^\alpha) \partial^\beta + x^\alpha (\partial^\sigma \partial^\beta)] - x^\rho [(\partial^\sigma x^\beta) \partial^\alpha + x^\beta (\partial^\sigma \partial^\alpha)] - x^\sigma [(\partial^\rho x^\alpha) \partial^\beta + x^\alpha (\partial^\rho \partial^\beta)] + x^\sigma [(\partial^\rho x^\beta) \partial^\alpha + x^\beta (\partial^\rho \partial^\alpha)] \}$$

$$= i^2 [g^{\alpha\sigma} x^\rho \partial^\beta + x^\rho x^\alpha \partial^\sigma \partial^\beta - g^{\rho\sigma} x^\rho \partial^\alpha - x^\rho x^\beta \partial^\sigma \partial^\alpha - g^{\rho\alpha} x^\sigma \partial^\beta - x^\sigma x^\alpha \partial^\rho \partial^\beta + g^{\rho\beta} x^\sigma \partial^\alpha + x^\sigma x^\beta \partial^\rho \partial^\alpha]$$

DOING (I) - (II) REMINDING THAT $g^{\mu\nu} = g^{\nu\mu}$, THE NETRIC IS SYMMETRIC

$$= i^2 \left\{ \underbrace{[g^{\rho\beta} x^\alpha \partial^\sigma + x^\alpha x^\rho \partial^\rho \partial^\sigma - g^{\sigma\beta} x^\alpha \partial^\rho - x^\alpha x^\sigma \partial^\beta \partial^\rho - g^{\rho\alpha} x^\beta \partial^\sigma - x^\beta x^\rho \partial^\alpha \partial^\sigma + g^{\sigma\alpha} x^\beta \partial^\rho + x^\beta x^\sigma \partial^\alpha \partial^\rho]}_{\text{I}} - \underbrace{[g^{\alpha\sigma} x^\rho \partial^\beta + x^\rho x^\alpha \partial^\sigma \partial^\beta - g^{\rho\sigma} x^\rho \partial^\alpha - x^\rho x^\beta \partial^\sigma \partial^\alpha - g^{\rho\alpha} x^\sigma \partial^\beta - x^\sigma x^\alpha \partial^\rho \partial^\beta + g^{\rho\beta} x^\sigma \partial^\alpha + x^\sigma x^\beta \partial^\rho \partial^\alpha]}_{\text{II}} \right\}$$

$$= i^2 [g^{\rho\beta} (x^\alpha \partial^\sigma - x^\sigma \partial^\alpha) - g^{\sigma\beta} (x^\rho \partial^\alpha - x^\alpha \partial^\rho) - g^{\rho\alpha} (x^\sigma \partial^\beta - x^\beta \partial^\sigma) + g^{\sigma\alpha} (x^\beta \partial^\rho - x^\rho \partial^\beta)]$$

$$= i (g^{\rho\beta} J^{\alpha\sigma} - g^{\sigma\beta} J^{\rho\alpha} - g^{\rho\alpha} J^{\sigma\beta} + g^{\sigma\alpha} J^{\beta\rho}) = [J^{\alpha\beta}, J^{\rho\sigma}]$$

MULTIPLYING BOTH SIDES BY "i" WE GOT

$$= -g^{\rho\beta} J^{\alpha\sigma} + g^{\sigma\beta} J^{\rho\alpha} + g^{\rho\alpha} J^{\sigma\beta} - g^{\sigma\alpha} J^{\beta\rho} = i [J^{\alpha\beta}, J^{\rho\sigma}]$$

I COULD USE THAT $J^{\mu\nu} = -J^{\nu\mu}$ TO CHANGE THESE MINUS

SIGNALS THAT APPEARS, BUT I PREFER NOT TO DO THIS 'CAUSE THIS

WAY LOOKS SIMILAR TO RAMOND'S RESULT.

b) One representation is given by $J_k^{\pm} = \frac{J_k \pm i K_k}{2}$.

We need this representation 'cause in the old one J closes but K is mixed so, writing in terms of plus and minus, we will be able to do a representation that is closed.

The old representation is

$$[J_i, J_j] = i \epsilon^{ijk} J_k$$

$$[K_i, K_j] = -i \epsilon_{ijk} J_k$$

$$[J_i, K_j] = i \epsilon_{ijk} K_k$$

Writing down the commutators in the new representation we got.

$$\begin{aligned} \text{I) } [J_i^+, J_j^+] &= \frac{1}{4} [(J_i + i K_i), (J_j + i K_j)] \\ &= \frac{1}{4} \{ [J_i, J_j] + i [J_i, K_j] + i [K_i, J_j] - [K_i, K_j] \} \\ &= \frac{1}{4} \{ \underbrace{i \epsilon_{ijk} J_k}_{\text{purple}} - \underbrace{\epsilon_{ijk} K_k}_{\text{blue}} + \underbrace{\epsilon_{ijk} K_k}_{\text{blue}} + \underbrace{i \epsilon_{ijk} J_k}_{\text{purple}} \} \\ &= \frac{1}{4} \{ i \epsilon_{ijk} (J_k + i K_k) + i \epsilon_{ijk} (J_k - i K_k) \} \\ &= \frac{1}{2} i \epsilon_{ijk} (J_k^+ + J_k^-) = \frac{i}{2} \epsilon_{ijk} J_k \end{aligned}$$

As we saw, all terms of K were canceled out in such way that we can rewrite this only in terms of J_k^+

$$= \frac{i}{2} \epsilon_{ijk} (2 J_k^+)$$

$$[J_i^+, J_j^+] = i \epsilon_{ijk} \underline{J_k^+}$$

$$\begin{aligned} \text{II) } [J_i^-, J_j^-] &= \frac{1}{4} [(J_i - i K_i), (J_j - i K_j)] \\ &= \frac{1}{4} \{ [J_i, J_j] - i [J_i, K_j] - i [K_i, J_j] - [K_i, K_j] \} \\ &= \frac{1}{4} \{ i \epsilon_{ijk} J_k + \epsilon_{ijk} K_k - \epsilon_{ijk} K_k + i \epsilon_{ijk} J_k \} \\ &= \frac{i}{2} \epsilon_{ijk} J_k \end{aligned}$$

But we want $[J_i^-, J_j^-] = i \epsilon_{ijk} J_k^- = \frac{i}{2} \epsilon_{ijk} (J_k - i K_k)$

As we saw the terms of K cancel out again and we can say that

$$[J_i^-, J_j^-] = i \epsilon_{ijk} \underline{J_k^-}$$

$$\begin{aligned}
 \text{III) } [J_i^-, J_j^+] &= \frac{1}{4} [(J_i - iK_i), (J_j + iK_j)] \\
 &= \frac{1}{4} \{ [J_i, J_j] + i[J_i, K_j] - i[K_i, J_j] + [K_i, K_j] \} \\
 &= \frac{1}{4} \{ i\epsilon_{ijk} J_k - \epsilon_{ijk} K_k + \epsilon_{ijk} K_k - i\epsilon_{ijk} J_k \} \\
 [J_i^-, J_j^+] &= 0 = [J_i^+, J_j^-]
 \end{aligned}$$

WE MAY ASSUME THAT IT IS AN IRREDUCIBLE REPRESENTATION 'CAUSE IT IS MADE WITH A CERTAIN OPERATION INVOLVING THE PREVIOUS REPRESENTATION. (THIS ALONE DOES NOT GUARANTEE THAT THIS REPRESENTATION IS AN IRREDUCIBLE ONE, BUT FOR OUR DEMONSTRATION I SUPPOSE IT IS SUFFICIENT)

$$2) a) \mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu}; \quad F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$$

$$\begin{aligned}
 \text{FIRST OF ALL WE MAY WRITE WHAT IS } -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} \\
 -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} &= -\frac{1}{4} (\partial_\mu A_\nu - \partial_\nu A_\mu) F^{\mu\nu} \\
 &= -\frac{1}{4} \partial_\mu A_\nu F^{\mu\nu} + \frac{1}{4} \partial_\nu A_\mu F^{\mu\nu}
 \end{aligned}$$

$$\begin{aligned}
 \text{THE INDEXES ARE MUTED AND } F^{\mu\nu} \text{ IS ANTISYMMETRIC } F^{\mu\nu} &= -F^{\nu\mu} \\
 &= -\frac{1}{2} \partial_\mu A_\nu F^{\mu\nu} = -\frac{1}{2} \partial_\mu A_\nu (\partial^\mu A^\nu - \partial^\nu A^\mu)
 \end{aligned}$$

WE CAN DOWN THE INDEXES USING THE METRIC

$$= -\frac{1}{2} \partial_\mu A_\nu g^{\mu\rho} g^{\nu\sigma} \underbrace{(\partial_\rho A_\sigma - \partial_\sigma A_\rho)}_{F_{\rho\sigma}}$$

$$-\frac{1}{4} F_{\mu\nu} F^{\mu\nu} = \frac{-g^{\mu\rho} g^{\nu\sigma}}{2} \partial_\mu A_\nu F_{\rho\sigma} = \mathcal{L} \underline{L_1}$$

THE EQUATION OF MOTION IS GIVEN BY EULER-LAGRANGE, THAT SAYS $\frac{\partial \mathcal{L}}{\partial \phi} - \partial_\mu \frac{\partial \mathcal{L}}{\partial \partial_\mu \phi} = 0$ WHERE $\phi = A_\mu$

FOR WE DO NOT HAVE CONFUSION I WILL CHANGE THE INDEXES

FOR α AND β IN THE E.L. EQUATION.

$$\frac{\partial \mathcal{L}}{\partial A_\beta} - \partial_\alpha \frac{\partial \mathcal{L}}{\partial \partial_\alpha A_\beta} = 0$$

AS WE CAN SEE OUR LAGRANGIAN DOES NOT DEPEND OF A_β

SO THE TERM $\frac{\partial \mathcal{L}}{\partial A_\beta} = 0$

$$\partial_\alpha \left[\frac{\partial}{\partial \partial_\alpha A_\beta} \frac{-g^{\mu\rho} g^{\nu\sigma}}{2} \partial_\mu A_\nu F_{\rho\sigma} \right] = 0$$

WE WILL HAVE A DERIVATIVE OF THE PRODUCT

$$\partial_\alpha \left[-\frac{g^{\mu\rho} g^{\nu\sigma}}{2} (\delta_\mu^\alpha \delta_\nu^\beta F_{\rho\sigma} + (\delta_\mu^\alpha \delta_\sigma^\beta - \delta_\sigma^\alpha \delta_\mu^\beta) \partial_\mu A_\nu) \right] = 0$$

$$\text{REMINING THAT } F_{\rho\sigma} = \partial_\rho A_\sigma - \partial_\sigma A_\rho$$

WE MAY IGNORE THE MINUS SIGNAL

$$\partial_\alpha \left[g^{\alpha\rho} g^{\beta\sigma} F_{\rho\sigma} + (g^{\alpha\mu} g^{\beta\nu} - g^{\alpha\nu} g^{\beta\mu}) \partial_\mu A_\nu \right] \frac{1}{2} = 0$$

$$\partial_\alpha \left[F^{\alpha\beta} + \underbrace{(\partial^\alpha A^\beta - \partial^\beta A^\alpha)}_{F^{\alpha\beta}} \right] \frac{1}{2} = 0$$

$$\partial_\alpha F^{\alpha\beta} = 0$$

b) By the NOETHER'S THEOREM A SCALAR FIELD TRANSFORMS AS

$$\phi'(x') = \phi(x), \text{ AS OUR SYMMETRY IS GIVEN BY } x'^\mu \rightarrow x^\mu + \alpha^\mu$$

WE WANT THE TRANSFORMATION IN THE SAME POINT

$$\phi(x) = \phi'(x) - \phi(x), \text{ AS } x'^\mu = x^\mu + \alpha^\mu \rightarrow x^\mu = x'^\mu - \alpha^\mu \Rightarrow x'^\mu = x^\mu - \alpha^\mu$$

CHANGING VARIABLES

$$\phi(x) = \phi(x^\mu - \alpha^\mu) - \phi(x^\mu)$$

WE MAY EXPAND $\phi(x^\mu - \alpha^\mu)$ UNTIL FIRST ORDER

$$\phi(x) = \cancel{\phi(x^\mu)} - \alpha^\mu \partial_\mu \phi(x^\mu) - \cancel{\phi(x^\mu)}$$

$$\phi(x) = -\alpha^\mu \partial_\mu \phi(x^\mu)$$

CONSIDERING A LAGRANGIAN $\mathcal{L} \equiv \mathcal{L}(\phi, \partial_\mu \phi)$ ITS VARIATION WILL

$$\text{be } \delta \mathcal{L} = \frac{\partial \mathcal{L}}{\partial \phi} \delta \phi + \frac{\partial \mathcal{L}}{\partial \partial_\mu \phi} \delta \partial_\mu \phi; \text{ AND } \delta \partial_\mu \phi = \partial_\mu \delta \phi$$

$$\text{WORKING OUT } \partial_\nu \delta \phi = \partial_\nu [-\alpha^\mu \partial_\mu \phi(x)] = -\alpha^\mu \partial_\nu \partial_\mu \phi(x^\mu)$$

$$\delta \mathcal{L} = \frac{\partial \mathcal{L}}{\partial \phi} \delta \phi + \frac{\partial \mathcal{L}}{\partial \partial_\mu \phi} \partial_\mu \delta \phi$$

$$\text{USING EULER-LAGRANGE EQUATION } \frac{\partial \mathcal{L}}{\partial \phi} = \partial_\mu \frac{\partial \mathcal{L}}{\partial \partial_\mu \phi}$$

$$\delta \mathcal{L} = \partial_\mu \frac{\partial \mathcal{L}}{\partial \partial_\mu \phi} \delta \phi + \frac{\partial \mathcal{L}}{\partial \partial_\mu \phi} \partial_\mu \delta \phi$$

THAT WE CAN SEE IS THE DERIVATIVE OF THE PRODUCT AND WE CAN

WRITE AS THE TOTAL DERIVATIVE

$$\delta \mathcal{L}_\phi = \partial_\mu \left[\frac{\partial \mathcal{L}}{\partial \partial_\mu \phi} \delta \phi \right]$$

WE ALSO NEED TO EXPAND THIS LAGRANGIAN IN TERMS OF x^μ

$$\delta \mathcal{L}_x = \frac{\partial \mathcal{L}}{\partial x^\mu} \delta x^\mu + \cancel{\frac{\partial \mathcal{L}}{\partial \partial_\mu x^\mu} \delta \partial_\mu x^\mu} \rightarrow \mathcal{L} \text{ DOESN'T DEPEND OF } \partial_\mu x^\mu$$

$$\text{DOING } \delta \mathcal{L}_\phi = \delta \mathcal{L}_X$$

$$\partial_\mu \left[\frac{\partial \mathcal{L}}{\partial \partial_\mu \phi} \delta \phi \right] = \frac{\partial \mathcal{L}}{\partial X^\mu} \delta X^\mu ; \text{ The NOTATION SAYS TO US } \frac{\partial}{\partial X^\mu} = \partial_\mu$$

$$\partial_\mu \left[\frac{\partial \mathcal{L}}{\partial \partial_\mu \phi} \delta \phi + \mathcal{L} \delta X^\mu \right] = 0 ; \text{ DOING } \delta \phi = -\alpha^\mu \partial_\mu \phi \text{ AND } \delta X = \alpha^\mu$$

$$\partial_\mu \left[\frac{\partial \mathcal{L}}{\partial \partial_\mu \phi} (-\alpha^\nu \partial_\nu \phi) + \mathcal{L} \alpha^\mu \right] = 0$$

$$-\alpha^\nu \partial_\mu \left[\frac{\partial \mathcal{L}}{\partial \partial_\mu \phi} \partial_\nu \phi - g^\mu_\nu \mathcal{L} \right] = 0$$

Where we know $\partial_\mu J^\mu = 0$ So, the conserved current will be

$$J^\mu = \frac{\partial \mathcal{L}}{\partial \partial_\mu \phi} \partial_\nu \phi - g^\mu_\nu \mathcal{L} \text{ THAT IS ALSO the ENERGY-MOMENTUM TENSOR}$$

$$T^\mu_\nu = \frac{\partial \mathcal{L}}{\partial \partial_\mu \phi} \partial_\nu \phi - g^\mu_\nu \mathcal{L} \Rightarrow T^\mu_\nu = -F^{\mu\nu} \partial_\nu \phi - g^\mu_\nu \mathcal{L}$$

3) FIRST we need to know if the transformation is a symmetry

$$X \rightarrow X' = e^\epsilon X$$

$$\phi(x) \rightarrow \phi'(x') = e^{-d_\phi \epsilon} \phi(x) ; \text{ where } d_\phi = 1 \text{ so}$$

$$\phi(x) \rightarrow \phi'(x') = e^{-\epsilon} \phi(x)$$

by NOETHER'S THEOREM we need to expand X'^μ AND $\phi'(x'^\mu)$

$$X'^\mu = (1 + \epsilon) X^\mu ; \text{ where I EXPANDED the EXPONENTIAL IN } \epsilon \ll 1$$

$$X'^\mu = X^\mu + \epsilon X^\mu \rightarrow X^\mu = X'^\mu - \epsilon X^\mu$$

USING THIS we MAY WRITE $\phi'(x)$ AS $\phi'(X'^\mu - \epsilon X^\mu)$ AND TO KNOW

ITS FACE we MAY TAYLOR EXPAND IT

$$\phi'(X'^\mu - \epsilon X^\mu) = \phi'(X'^\mu) - \epsilon X^\mu \partial_\mu \phi'(X'^\mu)$$

The Term $\phi'(x')$ IS GIVEN by the TRANSFORMATION THAT we CAN

EXPAND AND see THAT $\phi'(x') = (1 - \epsilon) \phi(x) ;$ USING IT

$$\phi'(X'^\mu - \epsilon X^\mu) = (1 - \epsilon) \phi(x) - \epsilon X^\mu \partial_\mu (1 - \epsilon) \phi(x)$$

$$= \phi(x) - \epsilon \phi(x) - \epsilon X^\mu \partial_\mu \phi + \cancel{\mathcal{O}(\epsilon^2)} \rightarrow \text{I will ignore these terms}$$

TO NOETHER'S THEOREM WHAT TRULY MATTERS TO US IS THE VARIATION OF

the FIELD AT THE SAME POINT

$$\delta \phi = \phi'(x) - \phi(x) ; \text{ we ALREADY WROTE } \phi'(x)$$

$$= \cancel{\phi(x)} - \epsilon \phi(x) - \epsilon X^\mu \partial_\mu \phi(x) - \cancel{\phi(x)}$$

$$\delta \phi = -\epsilon (\phi(x) + X^\mu \partial_\mu \phi(x))$$

TO see IF IT IS A SYMMETRY we HAVE TO LOOK IF the ACTION

REMAINS UNCHANGED i.e. $\delta S = 0$

before the transformation $S = \int d^4x \mathcal{L}$

When we apply the transformations into the Lagrangian we see

$$\partial_\mu = \frac{\partial}{\partial x^\mu} \quad \text{AND} \quad x'^\mu = e^\epsilon x^\mu \quad \text{by the transformation}$$

$$\text{So } \partial'_\mu = e^{-\epsilon} \partial_\mu$$

AND IS THE SAME FOR THE FIELD $\phi' = e^{-\epsilon} \phi$

THIS WAY THE TRANSFORMED KINETIC TERM OF THE LAGRANGIAN WILL BE

$$\frac{e^{-4\epsilon}}{2} \partial_\mu \phi \partial^\mu \phi$$

THE POTENTIAL PART WILL BE $e^{-4\epsilon} \frac{\lambda}{4!} \phi^4$ WHEN MODIFIED

AND THE INTEGRATION VARIABLE, MODIFIED, WILL BE $d^4x' = e^{4\epsilon} d^4x$

THIS WAY THE ACTION IS

$$S' = \int e^{4\epsilon} d^4x \left(e^{-4\epsilon} \frac{1}{2} \partial_\mu \phi \partial^\mu \phi - e^{-4\epsilon} \frac{\lambda}{4!} \phi^4 \right)$$
$$= \int d^4x \mathcal{L}$$

IT SHOWS US THAT $S = S' \rightarrow \delta S = 0$. IT IS A SYMMETRY.

TO SEE ITS CONSERVED CURRENT WE MAY WRITE THE NOETHER'S CURRENT EXPRESSION, THAT WE FOUND IN THE PREVIOUSLY QUESTION

$$J^\mu = \frac{\partial \mathcal{L}}{\partial \partial_\mu \phi} \delta \phi + \mathcal{L} \delta x^\mu$$

$$\text{TAKING } \mathcal{L} = \frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{\lambda}{4!} \phi^4$$

$$\frac{\partial \mathcal{L}}{\partial \partial_\mu \phi} = \partial^\mu \phi$$

$$\delta \phi = -\epsilon (\phi(x) + x^\mu \partial_\mu \phi(x))$$

$$\delta x^\mu = \epsilon x^\mu$$

SO THE CONSERVED CURRENT IS

$$J^\mu = -\epsilon \partial^\mu \phi (\phi + x^\nu \partial_\nu \phi) + \epsilon x^\mu \mathcal{L}$$
$$= -\epsilon \partial^\mu \phi \phi - \epsilon \partial^\mu \phi x^\nu \partial_\nu \phi + \epsilon x^\mu \mathcal{L}$$

THE ENERGY-MOMENTUM TENSOR IS $T^{\mu\nu} = \partial^\mu \phi \partial^\nu \phi - g^{\mu\nu} \mathcal{L}$

$$J^\mu = -\epsilon \left[\partial^\mu \phi \phi + \underbrace{\partial^\mu \phi x^\nu \partial_\nu \phi}_{= x_\nu \partial^\nu \phi} - x^\mu \mathcal{L} \right]$$

SO

$$J^\mu = -\epsilon \left[\partial^\mu \phi \phi + x_\nu \partial^\mu \phi \partial^\nu \phi - x^\mu \mathcal{L} \right]$$

IF WE TAKE OFF x_ν FROM THE SECOND AND THIRD TERMS WE WILL GET THE $T^{\mu\nu}$ THAT WAS WRITTEN

NOW WE YIELD

$$\mathcal{J}^\mu = -\epsilon [\partial^\mu \phi \phi + \chi_\nu T^{\mu\nu}]$$

TO KNOW IF THE SIGNAL IS CORRECT WE NEED TO TAKE THE

DIVERGENT OF $\mathcal{J}^\mu$; $\partial_\mu \mathcal{J}^\mu = 0$

$$\partial_\mu \mathcal{J}^\mu = \partial_\mu \{-\epsilon [\partial^\mu \phi \phi + \chi_\nu T^{\mu\nu}]\} ; \epsilon \text{ IS A CONSTANT}$$

$$= -\epsilon \partial_\mu [\partial^\mu \phi \phi + \chi_\nu T^{\mu\nu}]$$

$$= -\epsilon \{ [\partial_\mu (\partial^\mu \phi) \phi + \partial^\mu \phi \partial_\mu \phi] + [\partial_\mu (\chi_\nu) T^{\mu\nu} + \underbrace{\chi_\nu \partial_\mu (T^{\mu\nu})}_{=0}] \}$$

IF THE ENERGY IS CONSERVED $\partial_\mu T^{\mu\nu} = 0$

$$= -\epsilon [\square \phi \phi + (\partial_\mu \phi)^2 + \delta_{\mu\nu} T^{\mu\nu}]$$

'CAUSE $\partial_\mu \chi_\nu = \delta_{\mu\nu}$

$$= -\epsilon [\square \phi \phi + (\partial_\mu \phi)^2 + \overbrace{T^\mu_\mu}^{\text{TRACE}}]$$

$$\times \partial_\mu \left[\frac{\partial \mathcal{L}}{\partial \partial_\mu \phi} \right] - \frac{\partial \mathcal{L}}{\partial \phi} = 0$$

$$\partial_\mu [\partial^\mu \phi] + \frac{\lambda}{3!} \phi^3 = 0$$

$$\square \phi = -\frac{\lambda}{3!} \phi^3$$

$$\left\{ \begin{array}{l} \underline{T^\mu_\mu}: = (\partial^\mu \phi)^2 - 4 \left(\frac{1}{2} (\partial^\mu \phi)^2 - \frac{\lambda}{4!} \phi^4 \right) \\ = (\partial^\mu \phi)^2 - 2 (\partial^\mu \phi)^2 + \frac{\lambda}{6} \phi^4 \\ = -(\partial^\mu \phi)^2 + \frac{\lambda}{6} \phi^4 \end{array} \right.$$

$$= -\epsilon \left[-\frac{\lambda}{6} \phi^4 + (\partial_\mu \phi)^2 - (\partial^\mu \phi)^2 + \frac{\lambda}{6} \phi^4 \right]$$

$$= 0$$

THE SIGNAL IS CORRECT.

4) a) AN INFINITESIMAL LORENTZ INVARIATION IS GIVEN BY

$$X'^\mu = X^\mu + \delta \omega^\mu_\nu X^\nu, \text{ AND IT MIGHT TO BE INVARIANT.}$$

KNOWING THAT $X'^\mu = \Lambda^\mu_\nu X^\nu$ WE MAY WRITE $\Lambda^\mu_\nu = \delta^\mu_\nu + \delta \omega^\mu_\nu$

TO BE LORENTZ INVARIANT WE NEED A TRANSFORMATION THAT

LOOKS LIKE

$$g^{\mu\nu} = \Lambda^\alpha_\mu \Lambda^\beta_\nu g^{\alpha\beta} ; \text{ USING OUR APPROXIMATION}$$

$$g^{\mu\nu} = (\delta^\alpha_\mu + \delta \omega^\alpha_\mu) (\delta^\beta_\nu + \delta \omega^\beta_\nu) g^{\alpha\beta}$$

$$g^{\mu\nu} = \delta^\alpha_\mu \delta^\beta_\nu g^{\alpha\beta} + \delta^\alpha_\mu \omega^\beta_\nu g^{\alpha\beta} + \delta \omega^\alpha_\mu \delta^\beta_\nu g^{\alpha\beta} + \underbrace{\delta \omega^\alpha_\mu \delta \omega^\beta_\nu g^{\alpha\beta}}_{\text{THIS TERM MAY BE IGNORED CAUSE } \omega \text{ IS VERY SMALL}}$$

$$\cancel{g^{\mu\nu}} = \cancel{g^{\mu\nu}} + \delta \omega_{\mu\nu} + \delta \omega_{\nu\mu}$$

THIS WAY WE SEE THAT

$$\delta \omega_{\mu\nu} = -\delta \omega_{\nu\mu}$$

b) FOR ROTATIONS WE KNOW THAT TIME WON'T CHANGE, SO I MAY CHANGE NOTATION USING j, k FOR SPACE INDEXES AND THE GREEK ONES TO SPACE-TIME INDEXES.

WE KNOW FROM CLASSICAL MECHANICS THAT A ROTATION IS GIVEN BY $\delta \vec{X} = \vec{\theta} \times \vec{X}$, THAT REPRESENTS AN INFINITESIMAL ROTATION AROUND THE $\hat{\theta}$ DIRECTION.

IN NOTATION THE VECTOR PRODUCT IS GIVEN BY THE LEVI-CIVITA, SO WE MAY WRITE THIS AS

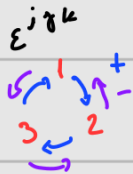
$$\delta X^j = \epsilon^{ijk} \theta^k X^j ; \text{ COMPARING WITH THE TERMS OF THE TRANSFORMATION } \delta X^\mu = \delta \omega^\mu_\nu X^\nu$$

WE SEE THAT

$$\delta \omega^j_k = \epsilon^{ijk} \theta^k ; \text{ WHERE I CHANGED } j \Leftrightarrow k \text{ 'CAUSE THEY ARE NOTED.}$$

IN A MATRIX THIS SHOULD BE

$$\delta \omega^j_k = \begin{pmatrix} 0 & \theta^3 & -\theta^2 \\ -\theta^3 & 0 & \theta^1 \\ \theta^2 & -\theta^1 & 0 \end{pmatrix}$$



c) AS WE HAVE DONE MANY TIMES BEFORE WE'RE GOING TO USE THE NOETHER'S CURRENT TO ANALYZE.

BUT BEFORE WE NEED TO KNOW HOW $\phi(x)$ CHANGES

NOETHER'S THEOREM TALKS ABOUT $\delta \phi$ IN THE SAME POINT OF SPACE

$$\delta \phi = \phi'(x) - \phi(x)$$

BUT WE NEED TO KNOW WHAT IS $\phi'(x)$

WE KNOW THAT $\phi'(x') = \phi(x + \delta \omega x)$ AND WE MAY TAYLOR EXPAND IT

$$\phi'(x') = \phi'(x) + \delta \omega^\mu_\nu X^\nu \partial_\mu \phi(x)$$

AND WE KNOW THAT $\phi'(x') = \phi(x)$ IS INVARIANT, SO

$$\phi(x) = \phi'(x) + \delta \omega^\mu_\nu X^\nu \partial_\mu \phi(x)$$

$$\underbrace{\phi'(x) - \phi(x)}_{\delta \phi} = - \delta \omega^\mu_\nu X^\nu \partial_\mu \phi(x)$$

$$\delta \phi = - \delta \omega^\mu_\nu X^\nu \partial_\mu \phi(x)$$

The NOETHER'S CURRENT IS GIVEN BY

$$J^\mu = \frac{\partial \mathcal{L}}{\partial \partial_\mu \phi} \delta \phi + \mathcal{L} \delta x^\mu$$

$$= -\frac{\partial \mathcal{L}}{\partial \partial_\mu \phi} \delta \omega^\mu_\nu x^\nu \partial_\mu \phi(x) + \mathcal{L} \delta \omega^\mu_\nu x^\nu$$

WE MAY USE THE SAME TRICK AS BEFORE TRYING TO WRITE IT IN

TERMS OF $T^{\mu\nu}$; $T^\mu_\nu = \frac{\partial \mathcal{L}}{\partial \partial_\mu \phi} \partial_\nu \phi - \delta^\mu_\nu \mathcal{L}$

I NEED TO UP THE INDEXES $\partial_\mu \phi$ AT THE FIRST TERM IN SUCH WAY THAT

$$= -\frac{\partial \mathcal{L}}{\partial \partial_\mu \phi} \delta \omega^\mu_\nu x^\nu g_{\mu\rho} \partial^\rho \phi(x) + \mathcal{L} \delta \omega^\mu_\nu x^\nu$$

$$= -\frac{\partial \mathcal{L}}{\partial \partial_\mu \phi} \delta \omega_{\rho\nu} x^\nu \partial^\rho \phi + \mathcal{L} \delta \omega^\mu_\nu x^\nu$$

PUTTING $-\delta \omega_{\rho\nu} x^\nu$ IN EVIDENCE

$$= -\delta \omega_{\rho\nu} x^\nu \left[\frac{\partial \mathcal{L}}{\partial \partial_\mu \phi} \partial^\rho \phi - \delta^{\mu\rho} \mathcal{L} \right]$$

$$= -\delta \omega_{\rho\nu} (x^\nu T^{\mu\rho} - x^\rho T^{\mu\nu}) \underline{h}$$

5) TO OBTAIN THE CONSERVED QUANTITIES I NEED TO APPLY THE NOETHER'S THEOREM

WE KNOW THAT THE FIELD CHANGES LIKE $A'^\mu(x') = \Lambda^\mu_\nu A^\nu(x)$

WE ALSO KNOW THAT WE MAY WRITE Λ IN ITS INFINITESIMAL FORM

$$\Lambda^\mu_\nu = \delta^\mu_\nu + \delta \omega^\mu_\nu$$

NOW WE ARE INTERESTED IN $\delta A^\mu = A'^\mu(x) - A^\mu(x)$

USING THAT $x'^\alpha = x^\alpha + \delta \omega^\alpha_\beta x^\beta \Rightarrow x^\alpha = x'^\alpha - \delta \omega^\alpha_\beta x^\beta$ I'LL BE ABLE TO WRITE

$A'(x)$ THAT IS SOMETHING I DON'T KNOW YET

$$A'(x) = A'(x'^\alpha - \delta \omega^\alpha_\beta x^\beta); \text{ WE MAY TAYLOR EXPAND IT}$$

$$= A'^\mu(x'^\alpha) - \delta \omega^\alpha_\beta x^\beta \partial_\alpha A'^\mu(x'); \text{ USING } A'^\mu(x') = \Lambda^\mu_\nu A^\nu(x)$$

$$= \Lambda^\mu_\nu A^\nu(x) - \delta \omega^\alpha_\beta x^\beta \partial_\alpha \Lambda^\mu_\nu A^\nu(x); \text{ EXPANDING } \Lambda \text{ TERMS}$$

$$= (\delta^\mu_\nu + \delta \omega^\mu_\nu) A^\nu(x) - \delta \omega^\alpha_\beta x^\beta \partial_\alpha (\delta^\mu_\nu + \delta \omega^\mu_\nu) A^\nu(x)$$

$$A'^\mu(x) = A^\mu(x) + \delta \omega^\mu_\nu A^\nu(x) - \delta \omega^\alpha_\beta x^\beta \partial_\alpha A^\mu; \text{ where I've IGNORED TERMS OF } \mathcal{O}(\omega^2)$$

$$\delta A^\mu = A'^\mu(x) - A^\mu(x) = \delta \omega^\mu_\nu A^\nu(x) - \delta \omega^\alpha_\beta x^\beta \partial_\alpha A^\mu$$

$$\delta A^\mu = \delta \omega^\mu_\nu A^\nu(x) - \delta \omega^\alpha_\beta x^\beta \partial_\alpha \underline{A^\mu}$$

$$\delta x^\mu = \delta \omega^\mu_\nu x^\nu \underline{h}$$

USING THE NOETHER'S CURRENT

$$J^\mu = \frac{\partial \mathcal{L}}{\partial \partial_\mu A_\sigma} \delta A_\sigma + \mathcal{L} \delta X^\mu ; \text{ WHEN WE DOWN THE INDEXES}$$

AND USING THAT $\mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu}$ WHERE $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$

WE SAW IN SECOND EXERCISE THAT $\frac{\partial \mathcal{L}}{\partial \partial_\mu A_\sigma} = -F^{\mu\sigma}$

SO WE NEED TO WRITE δA^μ AS δA_σ SO

$$\delta A_\sigma = g_{\sigma\mu} \delta A^\mu$$

$$\text{AND AS WE KNOW } \delta A^\mu = \delta \omega^\mu_\nu A^\nu(x) - \delta \omega^\alpha_\beta x^\beta \partial_\alpha A^\mu$$

$$\text{SO, } \delta A_\sigma = \delta \omega^\mu_\nu (\delta \omega^\mu_\nu A^\nu - \delta \omega^\alpha_\beta x^\beta \partial_\alpha A^\mu)$$

$$\delta A_\sigma = \delta \omega_{\sigma\nu} A^\nu - \delta \omega^\alpha_\beta x^\beta \partial_\alpha A_\sigma$$

WITH THIS

$$J^\mu = -F^{\mu\sigma} (\delta \omega_{\sigma\nu} A^\nu - \delta \omega^\alpha_\beta x^\beta \partial_\alpha A_\sigma) + \mathcal{L} \delta \omega^\mu_\nu x^\nu$$

$$J^\mu = -F^{\mu\sigma} \delta \omega_{\sigma\nu} A^\nu + F^{\mu\sigma} \delta \omega^\alpha_\beta x^\beta \partial_\alpha A_\sigma + \mathcal{L} \delta \omega^\mu_\nu x^\nu$$

TO FACTOR OUT $\delta \omega_{\sigma\nu}$ WE ARE GOING TO CHANGE THE INDEXES OF THE

SECOND FACTOR LIKE $\sigma \rightarrow \gamma$; $\beta \rightarrow \nu$, BECAUSE THEY ARE MUTED, AND

WE ARE GOING TO DOWN THEM WITH THE METRIC

$$\begin{aligned} J^\mu &= -F^{\mu\sigma} \delta \omega_{\sigma\nu} A^\nu + F^{\mu\gamma} g^{\nu\sigma} \delta \omega_{\sigma\nu} x^\nu \partial_\alpha A_\gamma + \mathcal{L} g^{\mu\sigma} \delta \omega_{\sigma\nu} x^\nu \\ &= -F^{\mu\sigma} \delta \omega_{\sigma\nu} A^\nu + \delta \omega_{\sigma\nu} \underbrace{(F^{\mu\gamma} \partial^\sigma A_\gamma + \mathcal{L} g^{\mu\sigma})}_{-T^{\mu\sigma}} x^\nu \end{aligned}$$

$$= -F^{\mu\sigma} \delta \omega_{\sigma\nu} A^\nu - \delta \omega_{\sigma\nu} T^{\mu\sigma} x^\nu$$

$$J^\mu = -\delta \omega_{\sigma\nu} (F^{\mu\sigma} A^\nu - F^{\mu\nu} A^\sigma + T^{\mu\sigma} x^\nu - T^{\mu\nu} x^\sigma) \quad \underline{\quad}$$

$$T^\mu_\nu = -F^{\mu\sigma} \partial_\nu A_\sigma - g^\mu_\nu \mathcal{L} \quad (\text{REMINDER})$$